

```

function [A, b] = matrixs(n)
    e = ones(n, 1);
    A = sparse(n, n);
    i = 1:n;
    a = spdiags([i'.*e], 0, n, n);
    a1 = spdiags([1/2*e], -1, n, n);
    a2 = spdiags([1/2*e], 1, n, n);
    a3 = spdiags([1/2*e], -2, n, n);
    a4 = spdiags([1/2*e], 2, n, n);
    A = a + a1 + a2 + a3 + a4;
    b = ones(n, 1);
end

n_values = [4, 10, 100, 1000, 10000, 100000];

for n = n_values
    [A, b] = matrixs(n);

    figure;
    spy(A);
    title(['Nonzero Matrix A for n = ', num2str(n)]);

    figure;
    spy(inv(full(A)));
    title(['Inverse of Matrix A for n = ', num2str(n)]);

    eigenvalues = eigs(A, 10, 'lm');
    figure;
    plot(real(eigenvalues), imag(eigenvalues), 'o');
    title(['Eigenvalues of Matrix A for n = ', num2str(n)]);

    % LU decomposition
    tic;
    [L, U, P] = lu(A);
    lu_time = toc;

    % Cholesky factorization
    tic;
    R = chol(A);
    chol_time = toc;

    fprintf('LU time for n = %d: %f seconds\n', n, lu_time);
    fprintf('Cholesky time for n = %d: %f seconds\n', n, chol_time);

    % Jacobi and Gauss-Seidel methods
    tol = 1e-8;
    max_iter = 1000;

    % Jacobi method
    x_jacobi = zeros(n, 1);

```

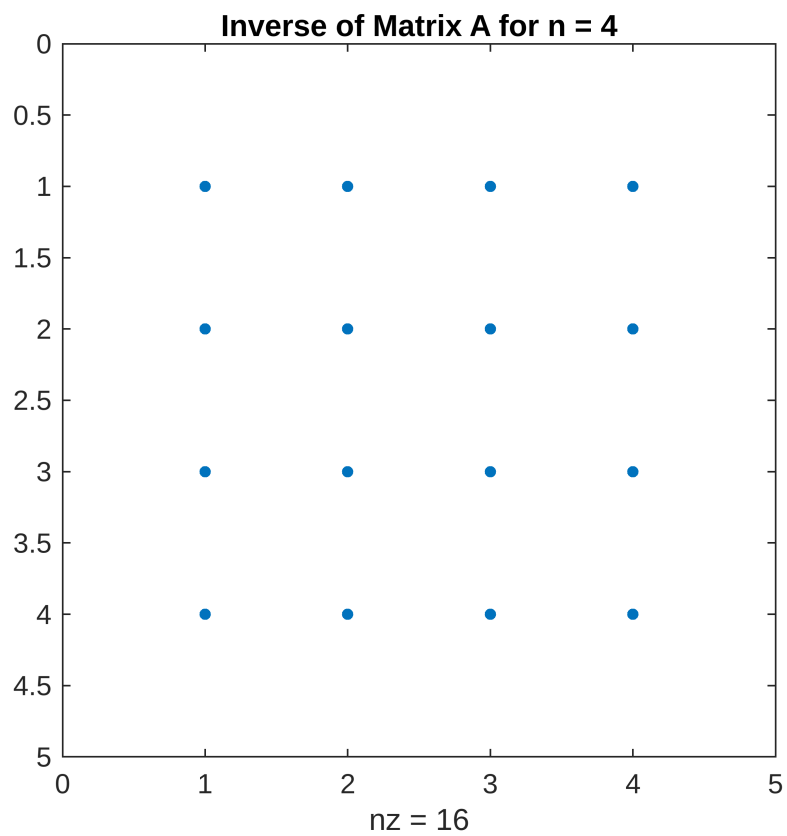
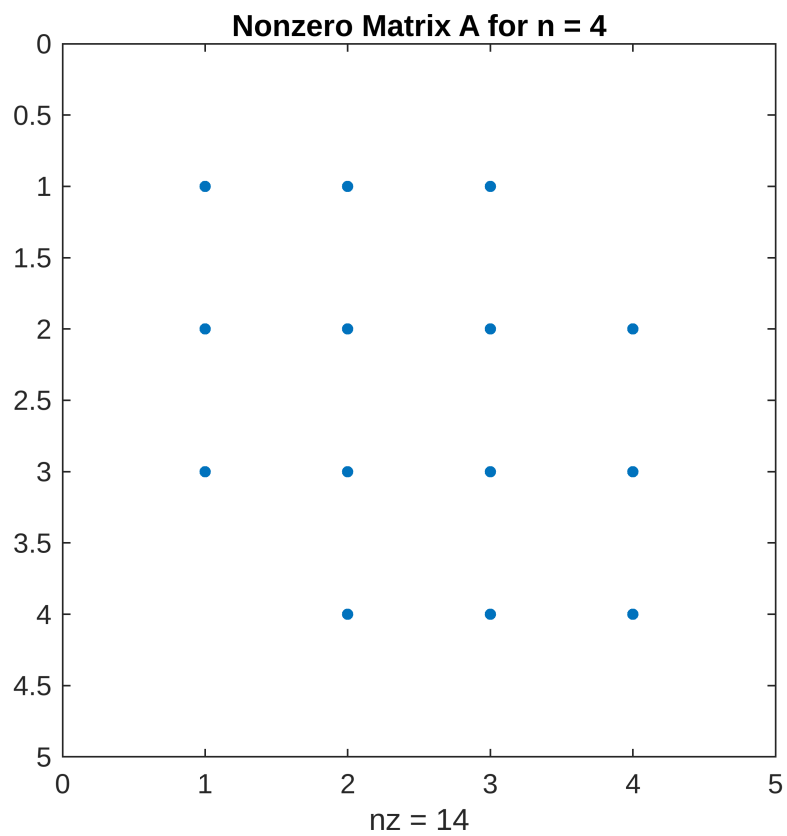
```

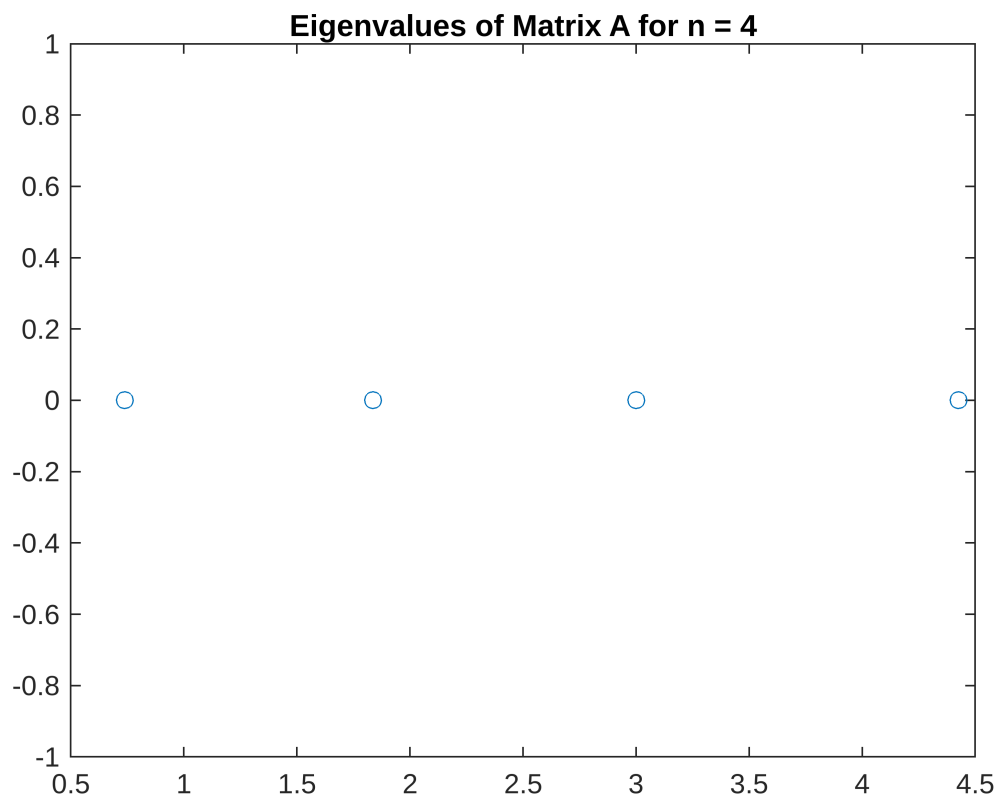
    for iter = 1:max_iter
        x_old = x_jacobi;
        for j = 1:n
            x_jacobi(j) = (b(j) - A(j, :) * x_old + A(j, j) * x_old(j)) /
A(j, j);
        end
        if norm(x_jacobi - x_old, 2) < tol
            break;
        end
    end

    % Gauss-Seidel method
    x_gs = zeros(n, 1);
    for iter = 1:max_iter
        x_old = x_gs;
        for j = 1:n
            x_gs(j) = (b(j) - A(j, :) * x_gs + A(j, j) * x_gs(j)) / A(j, j);
        end
        if norm(x_gs - x_old, 2) < tol
            break;
        end
    end

    fprintf('Jacobi method l2-norm for n = %d: %f, iterations: %d\n', n,
norm(x_jacobi, 2), iter);
    fprintf('Gauss-Seidel method l2-norm for n = %d: %f, iterations: %d\n',
n, norm(x_gs, 2), iter);
end

```



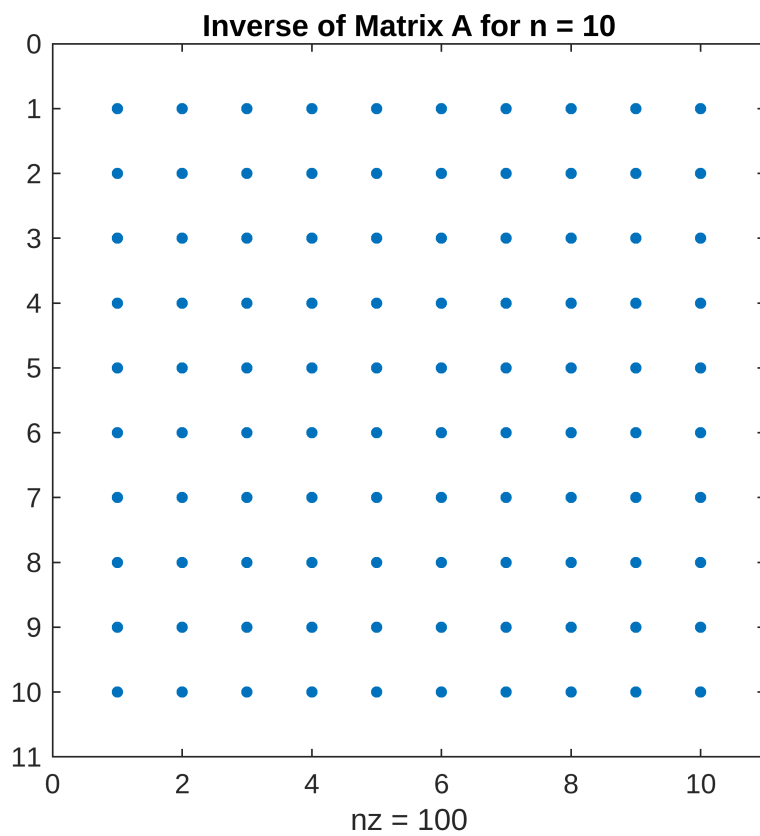
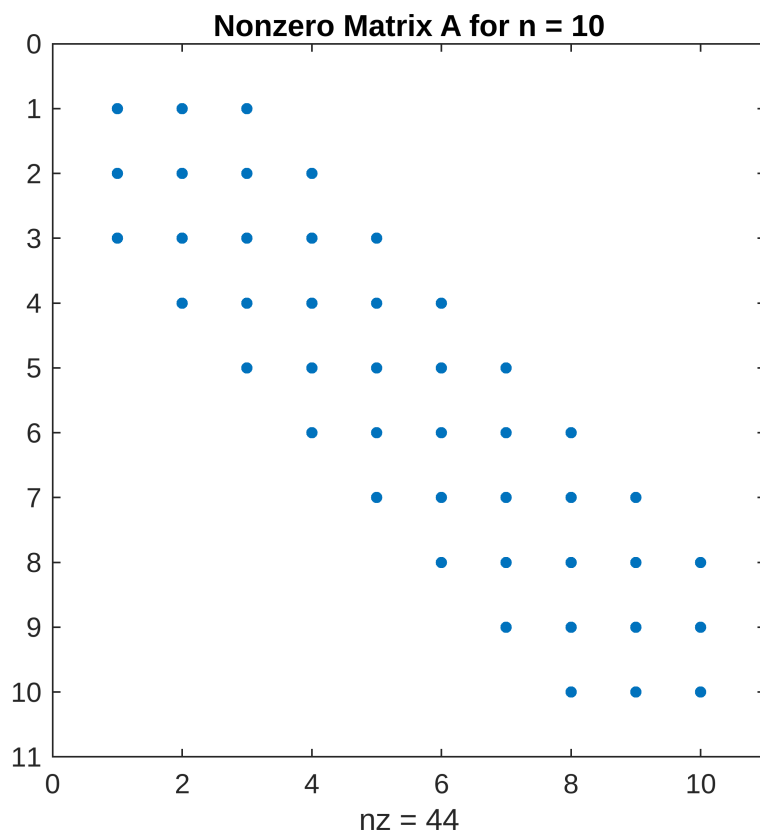


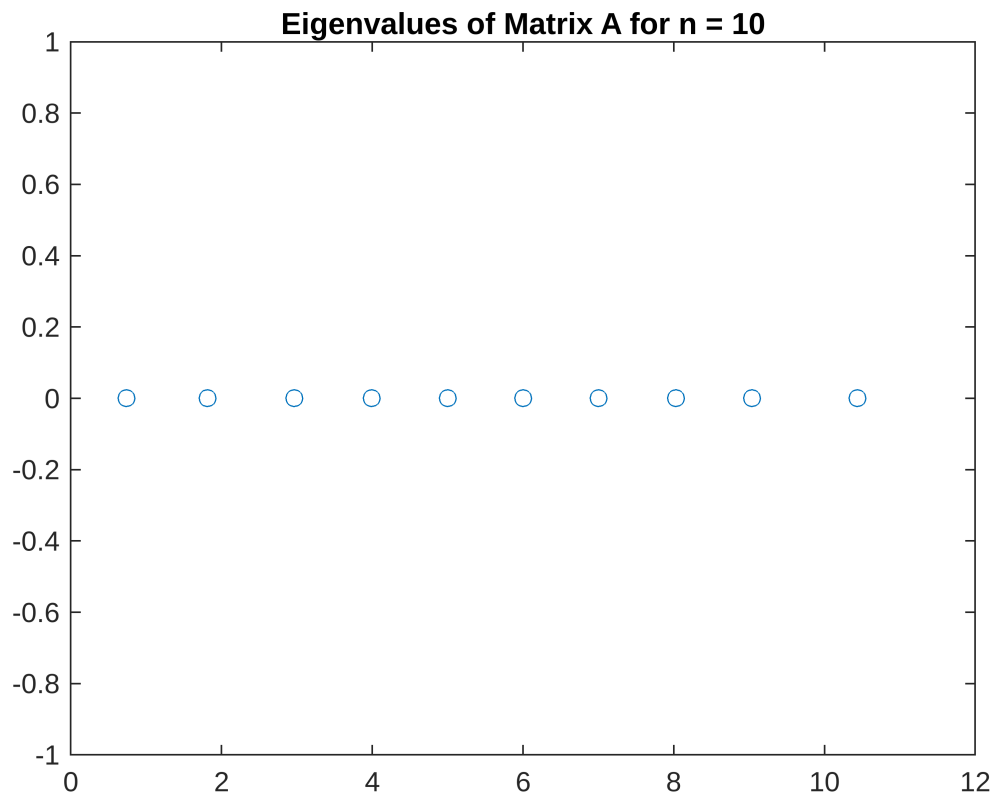
LU time for n = 4: 0.000093 seconds

Cholesky time for n = 4: 0.000058 seconds

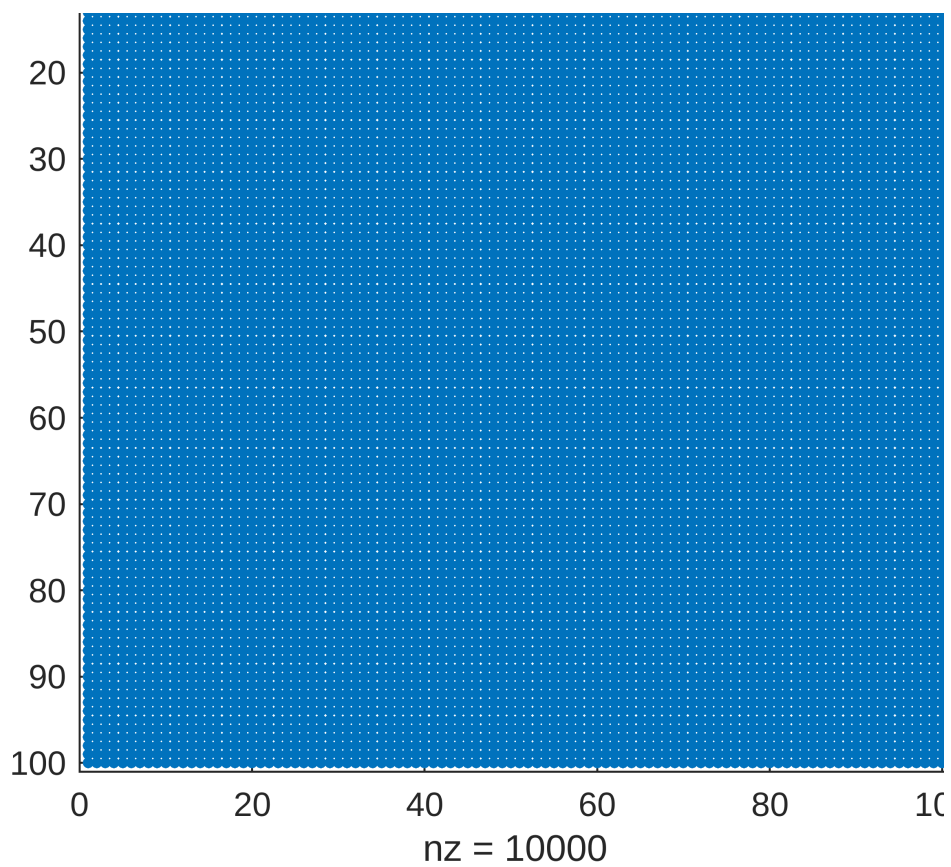
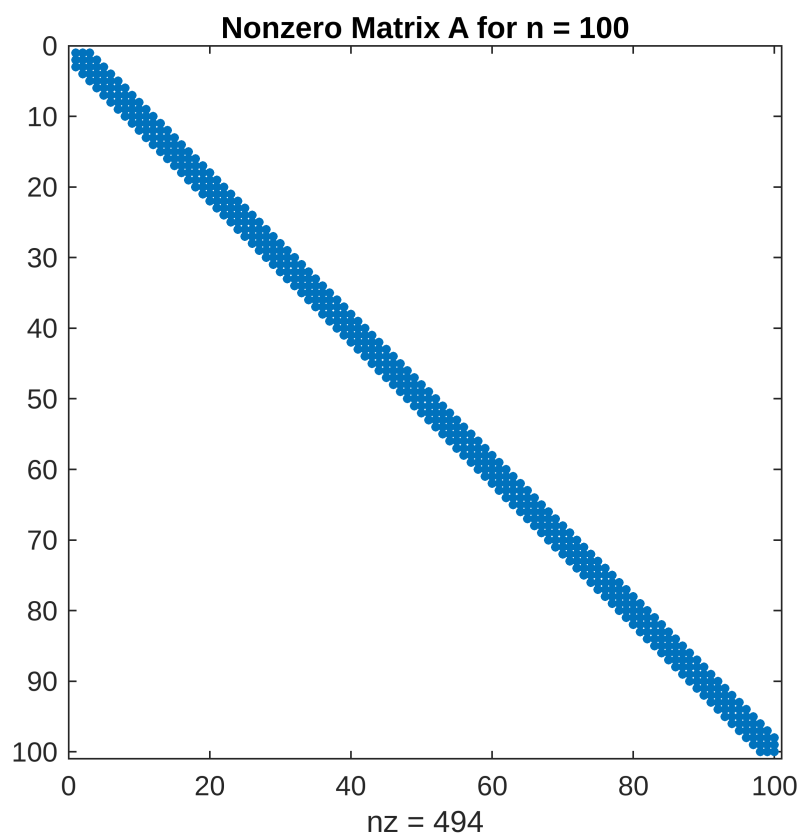
Jacobi method l2-norm for n = 4: 0.892679, iterations: 12

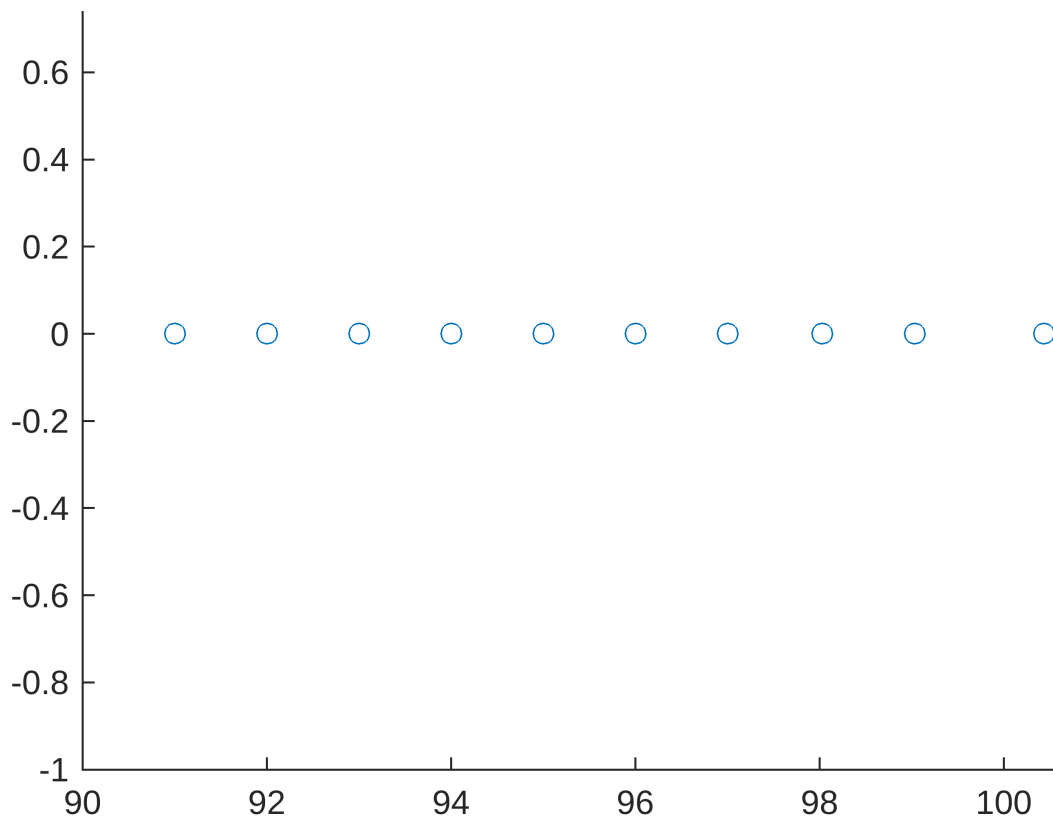
Gauss-Seidel method l2-norm for n = 4: 0.892679, iterations: 12



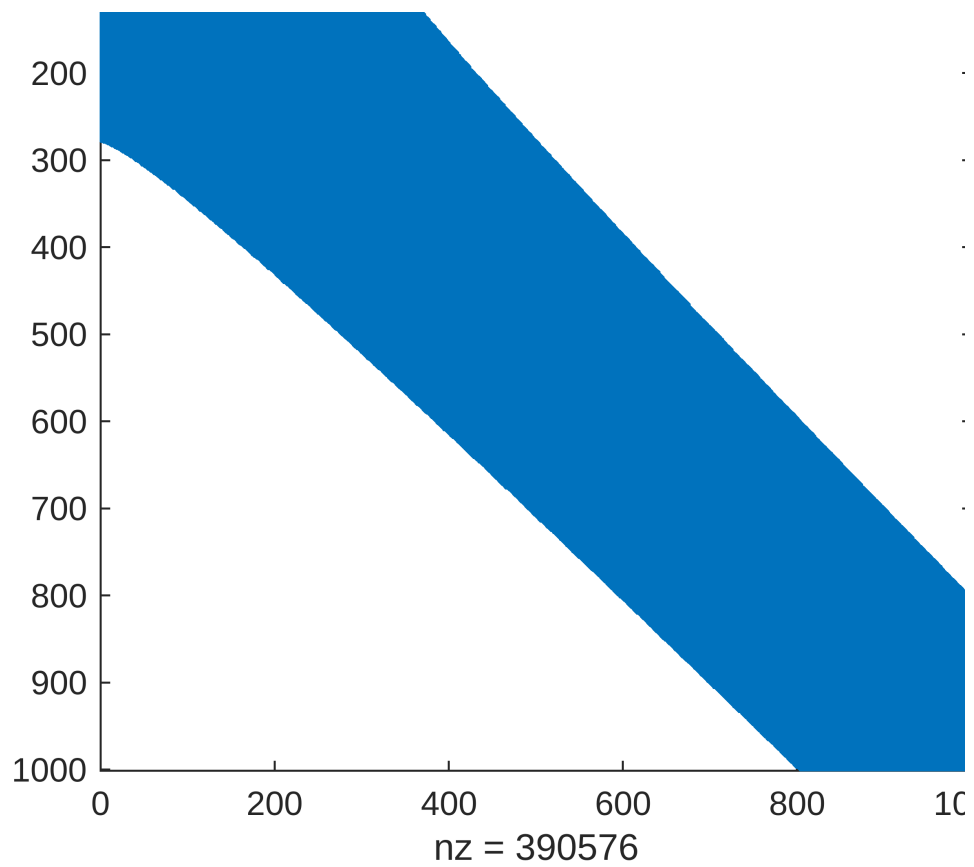
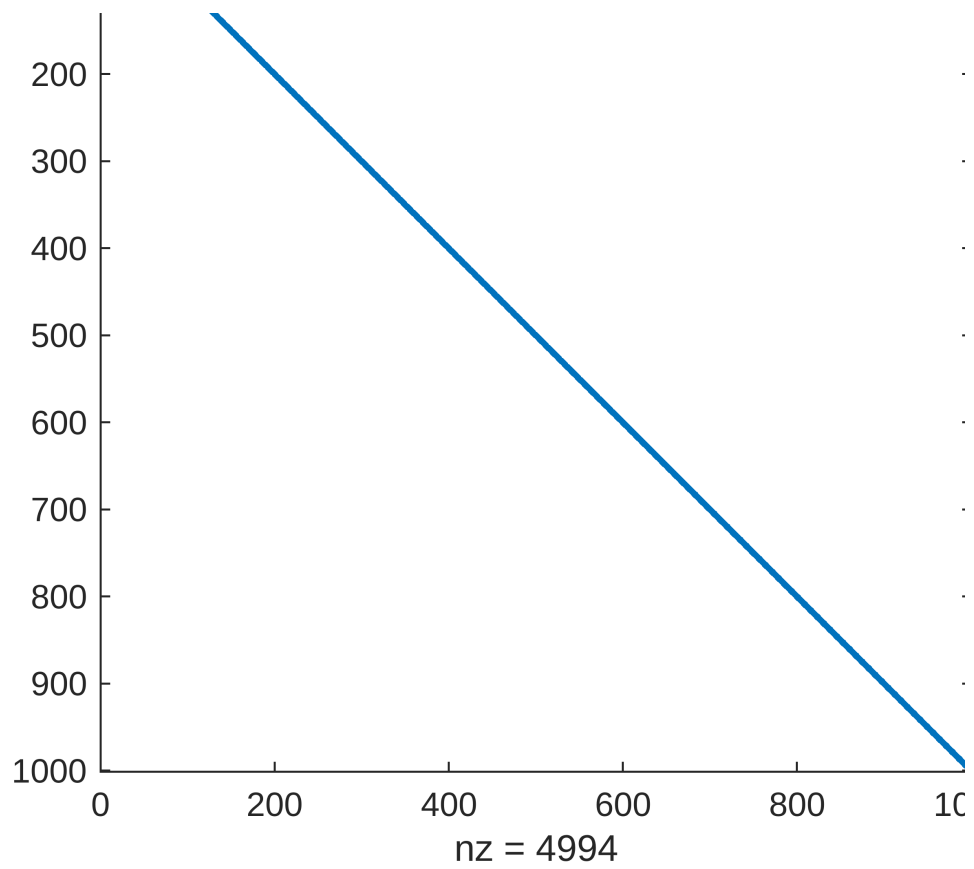


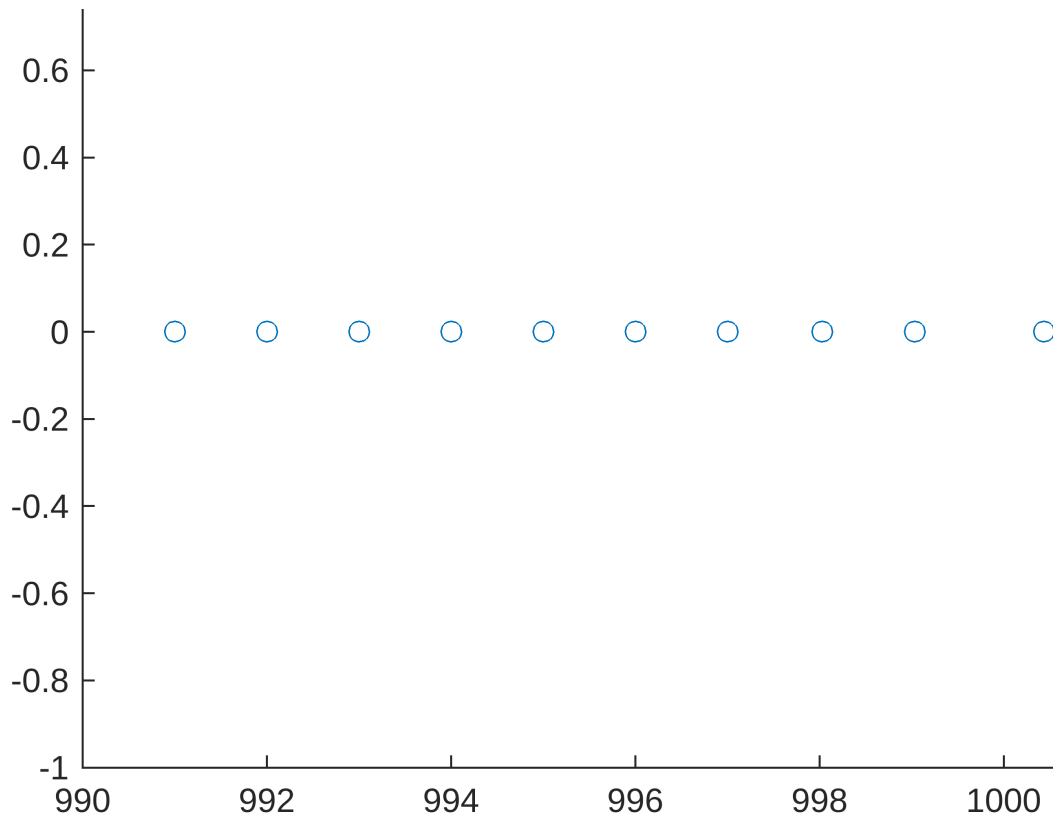
LU time for n = 10: 0.000092 seconds
Cholesky time for n = 10: 0.000057 seconds
Jacobi method l2-norm for n = 10: 0.932109, iterations: 11
Gauss-Seidel method l2-norm for n = 10: 0.932109, iterations: 11



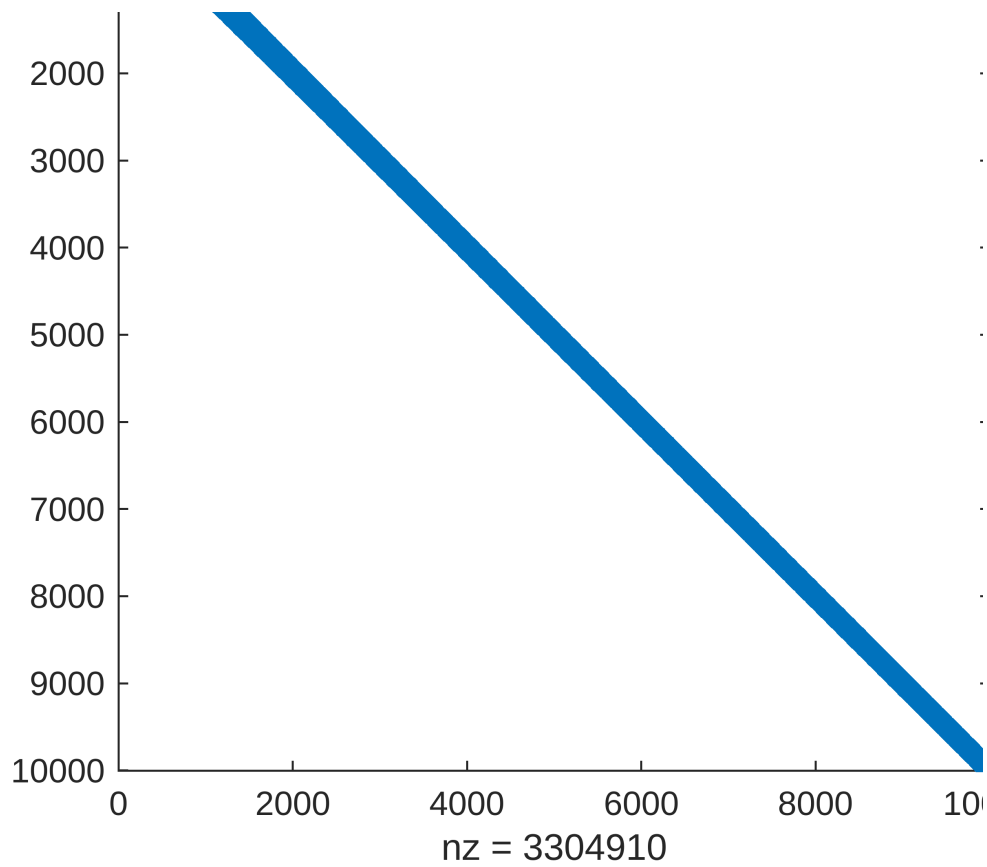
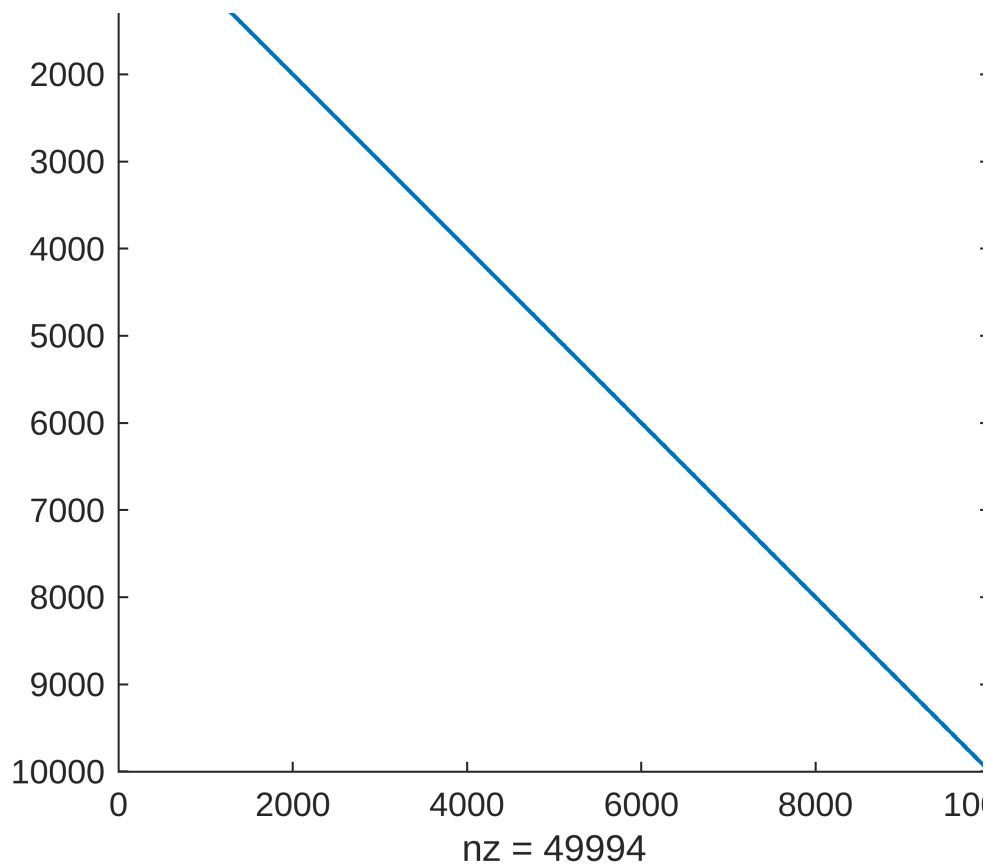


LU time for $n = 100$: 0.000113 seconds
Cholesky time for $n = 100$: 0.000066 seconds
Jacobi method l2-norm for $n = 100$: 0.968049, iterations: 11
Gauss-Seidel method l2-norm for $n = 100$: 0.968049, iterations: 11

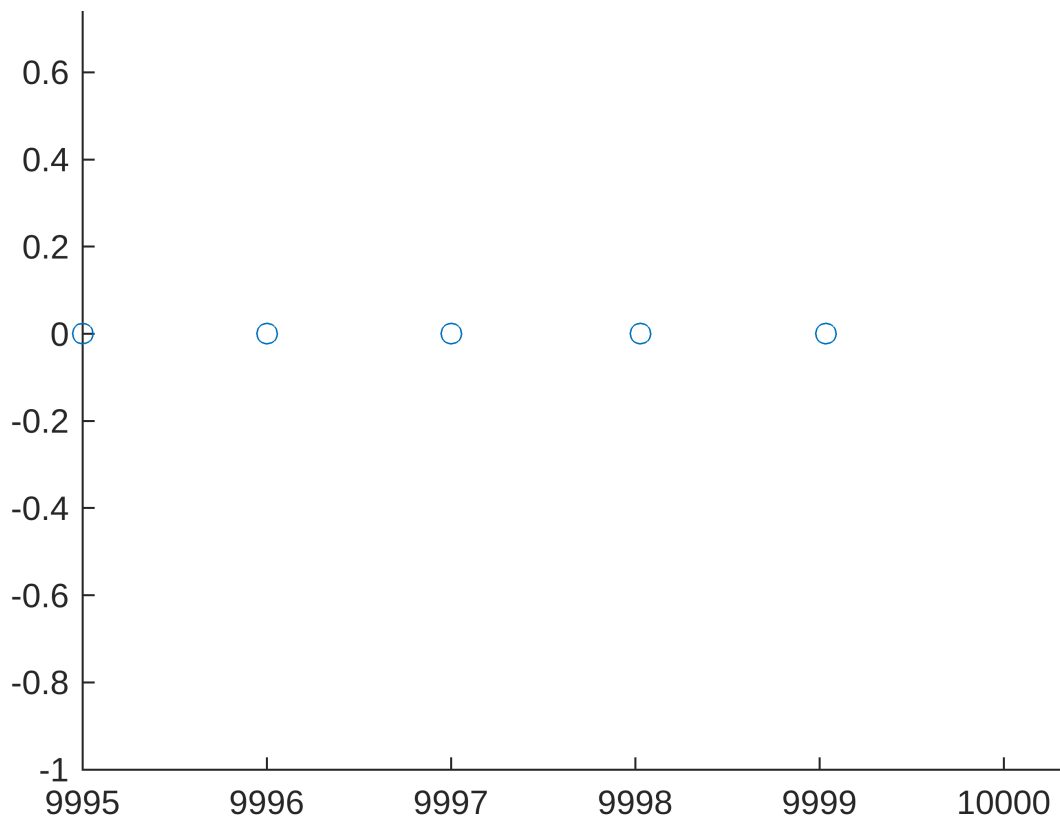




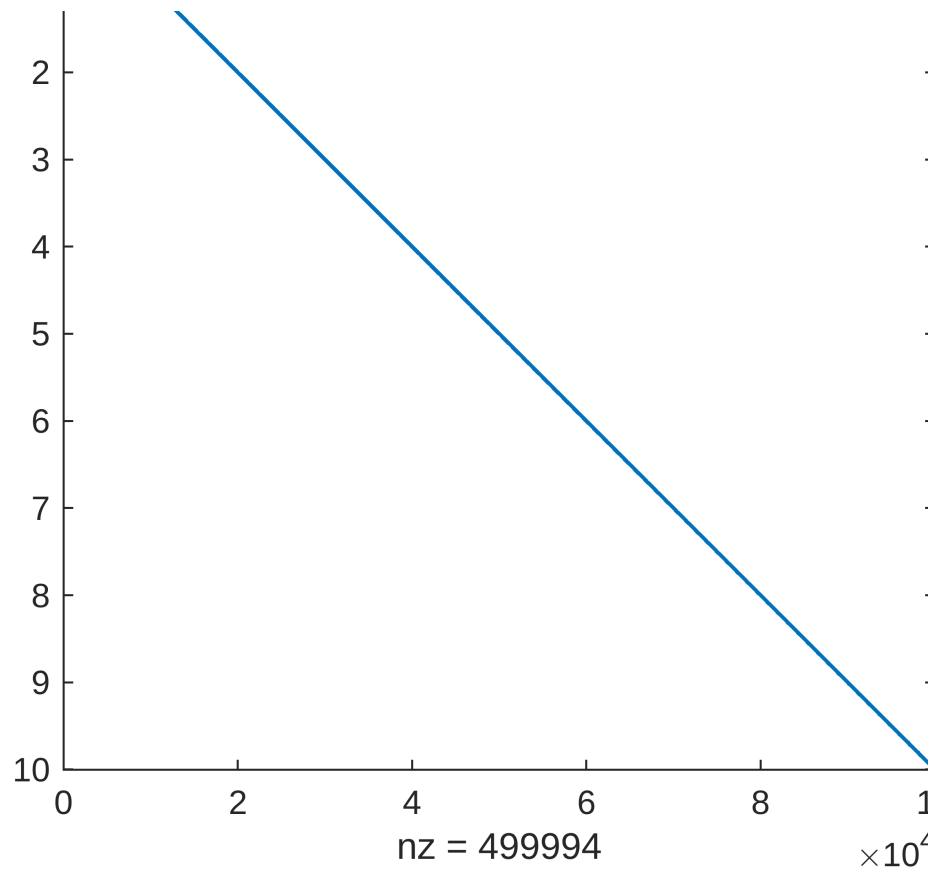
LU time for $n = 1000$: 0.000386 seconds
Cholesky time for $n = 1000$: 0.000252 seconds
Jacobi method l2-norm for $n = 1000$: 0.972561, iterations: 11
Gauss-Seidel method l2-norm for $n = 1000$: 0.972561, iterations: 11



Warning: 6 of the 10 requested eigenvalues converged. Eigenvalues that did not converge are NaN.



LU time for $n = 10000$: 0.003092 seconds
Cholesky time for $n = 10000$: 0.001903 seconds
Jacobi method l2-norm for $n = 10000$: 0.973022, iterations: 11
Gauss-Seidel method l2-norm for $n = 10000$: 0.973022, iterations: 11



Error using full
Requested 100000x100000 (74.5GB) array exceeds maximum array size preference (5.0GB). This might cause MATLAB to become unresponsive.

Related documentation